# LORETO GIRLS 2025 PREDICTION

## (EXPECTED KCSE EXAM)

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**SCHOOL……………………………………………….. INDEX NO………………………….**

**SIGN………………………………………………...**

***Kenya Certificate of Secondary Education (K.C.S.E)***

**233/1**

# CHEMISTRY

**PAPER 1**

**THEORY**

**TIME: 2 HOURS**

**INSTRUCTIONS TO CANDIDATES:**

*(i) Write your **name** and **index number** in the spaces provided **above**.*

*(ii) **Sign** and write the **date** of examination in the spaces provided **above**.*

*(iii) Answer **ALL** the questions in the spaces provided.*

*(iv) Mathematical tables and silent electronic calculators **may be** used.*

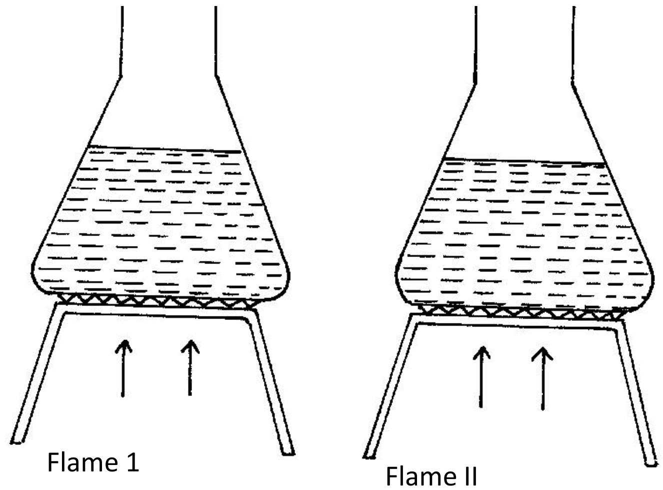
*(v) All working **must be** clearly shown where necessary.*

*(vi) Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing*

**FOR EXAMINER'S USE ONLY**

| Questions | Maximum Score | Candidate's Score |
|---|---|---|
| 1 –30 | 80 | |

**1.** The samples of equal volumes of water were put in 100cm$^3$ conical flasks and heated for 5 minutes on a Bunsen flame. It was observed that sample 1 registered a low temperature than sample II




(a) Name flame I **(1mk)**

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(b) State one disadvantage of using flame I for heating **(1mk)**

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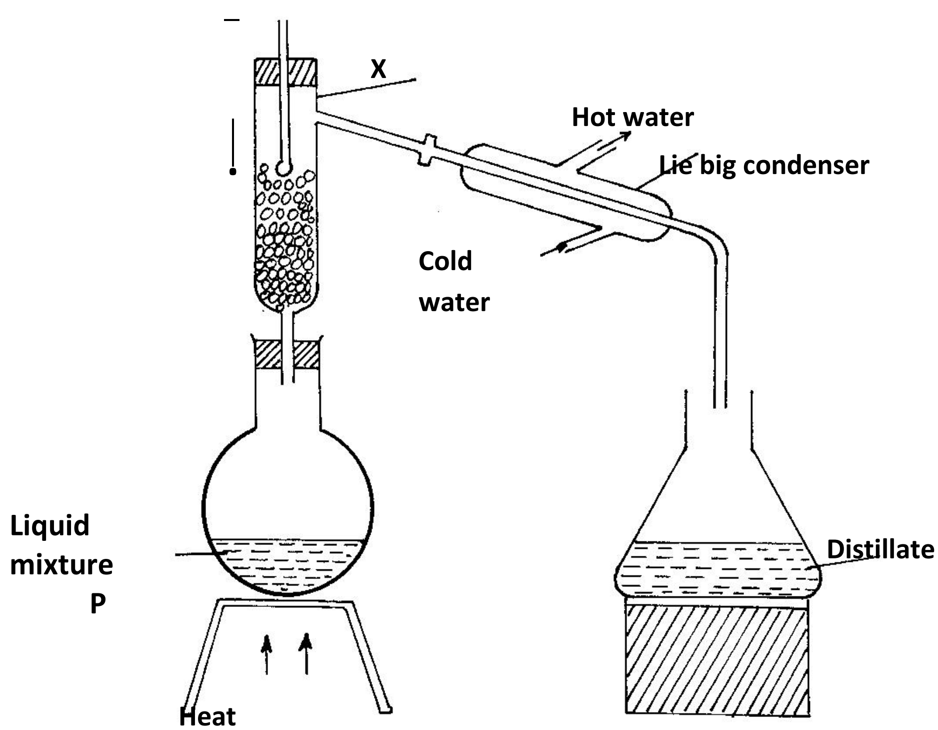
**2.** Study the diagram below and answer the questions that follow.

The diagram shows the method used to separate component of mixture P



(a) Name X **(1mk)**

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(b) What is the name given to the method used in separation of mixture P **(½mk)**

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(c) What would happen if the inlet and outlet of water were interchanged **( ½mk)**

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(d) Which physical property is used to separate mixture P **(1mk)**

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**3.** The table below shows the solubility of three solids P, Q, and R.

| **Solid** | **Cold Water** | **Hot Water** |
|---|---|---|
| P | soluble | soluble |
| Q | insoluble | insoluble |
| R | Insoluble | soluble |

How would you obtain pure samples of R,P and Q **(2mks)**

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**4.** State why a water molecule $H_2O$ can combine with $H^+$ ion to form $H_3O^+$ ion **(1mk)**

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**5.** The $P^H$ values of some solutions are given below

| **$P^H$** | 14.0 | 1.0 | 8.0 | 6.5 | 7.0 |
|---|---|---|---|---|---|
| **Solution** | M | L | N | P | Z |

(a) Identify the solution with the lowest concentration of hydrogen ion. Give reason for your answer **(1mk)**

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(b) Which solution would be used as an anti-acid for treating stomach upset. Give for your answer **(1mk)**

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**6.** The data below gives the electronic configuration of some selected atoms and ions

| Atom/ion | $A^{2+}$ | B | $C^{2-}$ | $D^{2+}$ | E | $F^{-}$ | $G^{+}$ | H |
|---|---|---|---|---|---|---|---|---|
| **Electronic configuration** | 2 | 2.4 | 2.8 | 2.8.8 | 2.8 | 2.8.8 | 0 | 2.8.2 |
| | | | | | | | | |

(a) Select an atom that is a noble gas **(1mk)**

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(b) What is the atomic number of C and A **(1mk)**

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(c) Select an element that belong to group 2 and period four **(1mk)**

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(d) Write the formula of the compound formed when D and F react **(1mk)**

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**7.** Helium is used instead of hydrogen in balloons for metrological research. **Explain (1mk)**

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**8.** Zinc metal and hydrochloric acid reacts according to the following equation

$$Zn_{(s)} + 2HCl_{(aq)} \longrightarrow ZnCl_{2\,(aq)} + H_{2\,(g)}$$

1.96g of Zinc metal were reacted with $100cm^3$ of 0.2M hydrochloric acid

a) Determine the reagent that was in excess **(2mks)**

Zn=65.2; Molar gas volume at s.t.p 22.4 liters

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(b) Calculate the total volume of hydrogen gas that was liberated at s.t.p **(1mk)**

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**9.** Give the IUPAC names of the following compounds **(1mk)**

(i) .$CH_3CH_2CH_2\underset{\displaystyle CH_3}{\underset{|}{C}}H\ CH_3$ ……………………………………………………………………………

(ii) $CH_3CH{=}CHCl$ …………………………………………………………………… **(1mk)**

**10.** 0.9g of potassium chloride and potassium carbonate mixture completely reacted with 25cm$^3$ of 0.2M hydrochloric acid

(i) Write an equation of the reaction which takes place **(1mk)**

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(ii) Determine the number of moles of the acid used **(1mk)**

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(iii) Calculate the mass of potassium chloride in the mixture (K=39.0; C=12.0; O=16.0) **(2mks)**

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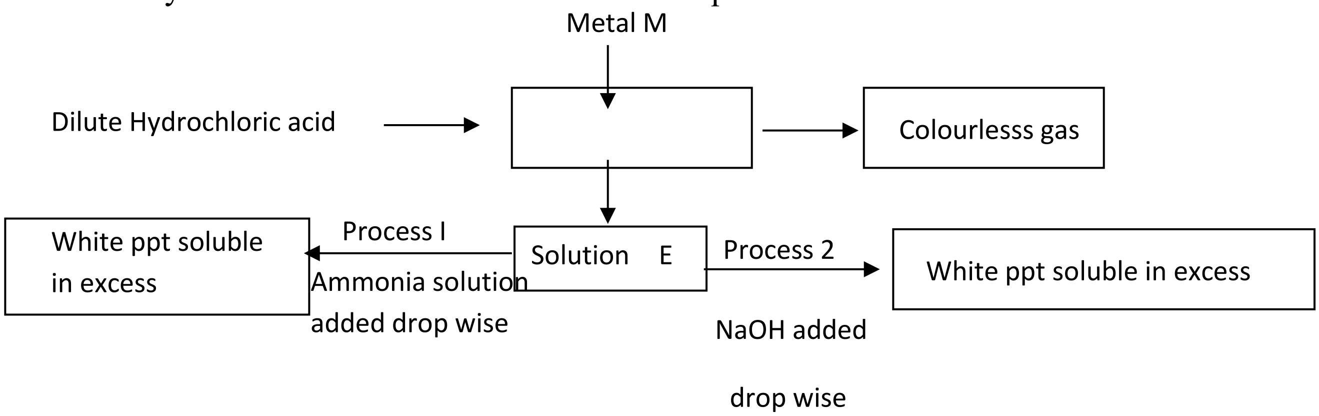
**11.** Study the flow chart below and answer the questions that follow




(i) Identify metal M: ……………………………………………………… **(1mk)**

(ii) Colourless gas: ……………………………………………………… **(1mk)**

(iii) Write an equation that leads to the formation of white precipitate in process

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**12.** a) Define the term dynamic equilibrium **(1mk)**

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b) A reaction at equilibrium can be represented as

$2CrO^{2-}{}_{4[aq]}$ $+2H^+[aq]$ $\rightleftharpoons$ $Cr_2O7\ {}^{2-}{}_{[aq]} + H_2O_{\{l\}}$

Yellow orange

State and explain the observation made when NaOH is added to the equilibrium mixture **(2mks)**

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13. Few drops of hydrochloric acid were added into a test tube containing lead {II} Nitrate solution

a) State one observation **made**

**(1mk)**

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b) Write an ionic equation of the reaction that occurred in the test **tube**

**(1mk)**

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**14.** A compound of carbon, hydrogen and oxygen contains 57.15% carbon, 4.76% hydrogen and the rest oxy gen. If its relative molecular mass is 126, find its molecular formula. (**C = 12, H = 1, O = 16)**

**(3mks)**

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**15.** Study the information in the table below and answer the questions that follow.

| Salt | Solubility g/100g of water | |
|---|---|---|
| | At 40°C | At 60°C |
| $CuSO_4$ | 28 | 38 |
| $Pb(NO_3)_2$ | 79 | 98 |

A mixture containing 35g of $CuSO_4$ and 78g of $Pb(NO_3)_2$ in 100g of water at 60°C was cooled to 40°C.

i) Which salt crystallized out? Give a reason.

**(2 marks)**

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ii) Calculate the mass of the salt that crystallized out.

**(1 mark)**

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**16.** a) Distinguish between strong and concentrated acid **( 1mk)**

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b). A solution of ammonia in methylbenzene has no effects on red litmus paper while a solution of ammonia in water turns red litmus paper blue. Explain **(2mks)**

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**17.** Name the process which takes place when

**i.** Iodine changes directly from solid to **gas** **(1mk)**

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**ii.** $Fe^{2+}_{(aq)}$ changes to $Fe^{3+}_{(aq)}$ **(1mk)**

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iii. White sugar changes to black when mixed with concentrated sulphuric (VI) acid (1mk)

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**18.** In the last stage of the solvay process, a mixture of sodium hydrogen carbonate and ammonium chloride is formed

**a)** State the method of separation used **(1mk)**

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**b)** Write an equation showing how lime is slaked **(1mk)**

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**c)** Name the by- product recycled in the above process **(1mk)**

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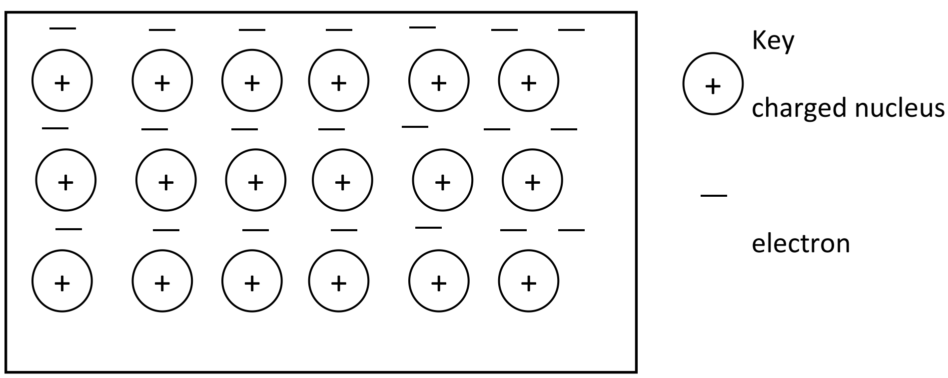
**19.** The diagram below is a section of a model of the structure of element K




**a)** State the type of bonding that exist in K **(1mk)**

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**b)** In which group of the periodic table does element K belong. Give a reason **(2mks)**

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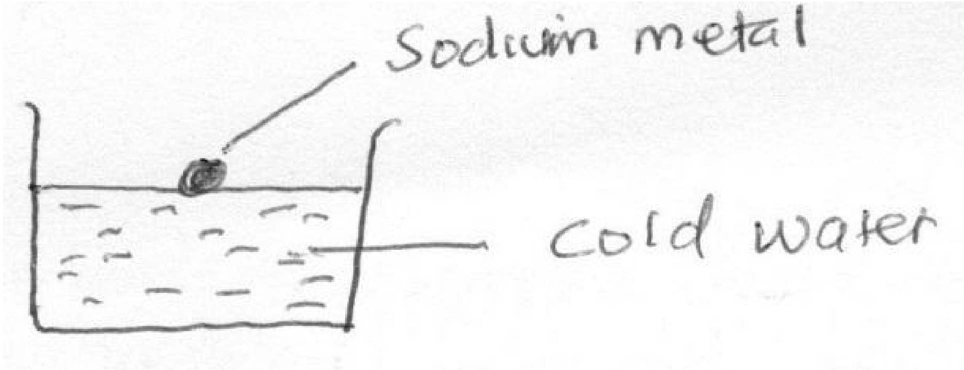
**20.** Study the diagram below and answer the questions that follow




**a)** State two observations made in the above experiment when sodium react with water **(2 mks)**

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**b)** Write a chemical equation for the reaction that takes **place (1mk)**

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**21.** (a) Explain why permanent hardness in water cannot be removed by boiling **(2mks)**

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(b) Name two methods that can be used to remove permanent hardness from water **(1mk)**

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**22.** Write an equation to show the effect of heat on the nitrate of: - **(2mks)**

i) Potassium

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(ii) Silver

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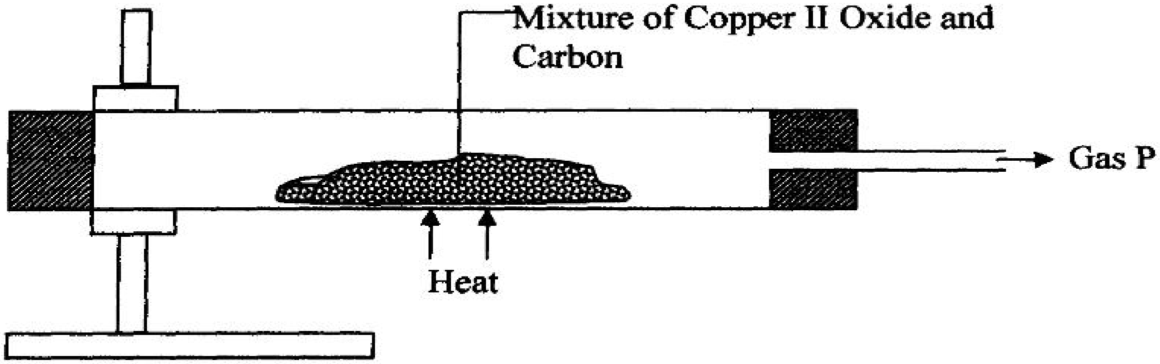
**23.** Study the diagram below and use it to answer the questions that follow.




**(a)** State the observation made in the combustion tube. **(1mk)**

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**(b)** Write an equation for the reaction that took place in the combustion tube. **(1mk)**

…………………………………………………………………………………………………………… (c)

Name gas **P** **(1mk)**

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**24.** Sulphur exists in two crystalline forms.

a) Name **one** crystalline form of Sulphur. **(1mk)**

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b) State **two** uses of Sulphur. **(2mks)**

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**25.** Bond energies for some bonds are tabulated below: -

| **BOND** | **BOND ENERGY KJ/mol** |
|---|---|
| H – H | 436 |
| C = C | 610 |
| C- H | 410 |
| C – C | 345 |

Use the bond energies to estimate the enthalpy for the reaction. **(3mks)**

$$C_2H_{4(g)} + H_{2(g)} \longrightarrow C_2H_{6(g)}$$

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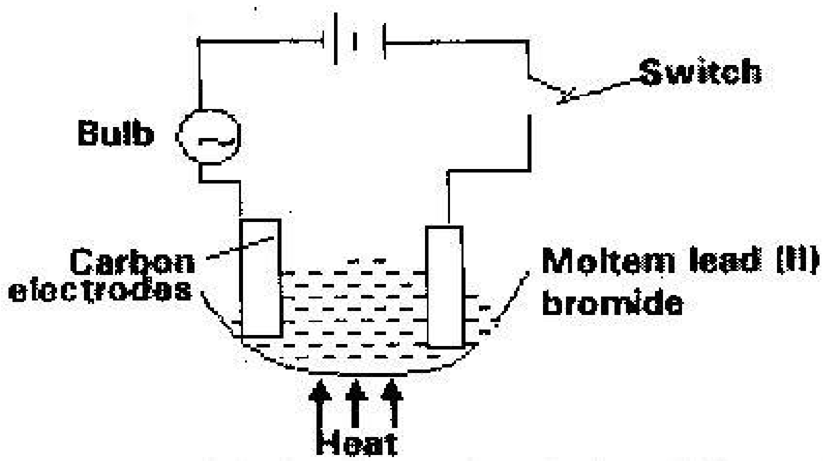
**26.** Study the set up below and answer the questions that flows




State all the observations that would be made when the circuit is completed

**(3mks)**

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**27.** Describe how solid samples of salts can be obtained from a mixture of lead (II) chloride, sodium chloride and ammonium chloride. **(3mks)**

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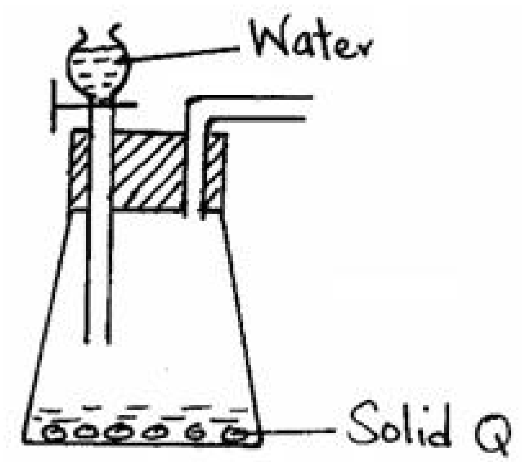
**28.** The diagram below represents a set-up used to prepare oxygen gas.




**(a)** Name substance Q. **(1mk)**

……………………………………………………………………………………………………..

**(b)** Complete the set-up to show how oxygen gas is collected. **(1mk)**

**(c)** Write the equation for the reaction that occur. **(1mk)**

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**29.** Two reagents that can be used to prepare chlorine gas are potassium manganate (VII) and hydrochloric acid.

(a) Write an equation for the reaction. **(1mk)**

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b) Give the formula of another reagent that can be used instead of potassium manganate (VII). **(1mk)**

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**(c)** Using an equation illustrate how chlorine bleach coloured substances. **(2mks)**

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